

LP-2 Salesforce Practical — Finalized Steps

Assignment: Student Record Management System

AIM

Develop a Student Record Management System using:

- Custom Object
- Apex Programming
- Visualforce Page

The application should:

- store student records
 - display records
 - perform basic insert operation
-

STEP 1 — Login to Salesforce

Open:

[Salesforce Developer Login](#)

Login to your Developer Org.

STEP 2 — Create Custom Object

Go to:

Setup → Object Manager → Create → Custom Object

Fill:

Field	Value
Label	Student
Plural Label	Students
Record Name	Student Name
Data Type	Text

Click:

Save

STEP 3 — Create Custom Fields

Go to:

Object Manager → Student → Fields & Relationships → New

Create these fields:

Field 1 — Roll No

Property	Value
Data Type	Number
Field Label	Roll No
Required	Yes

API Name auto becomes:

Roll_No__c

Click Save.

Field 2 — Class

Property	Value
Data Type	Text
Field Label	Class

API Name:

Class__c

Click Save.

Field 3 — Mobile No

Property	Value
Data Type	Phone
Field Label	Mobile No

API Name:

Mobile_No__c

Click Save.

STEP 4 — Create Apex Controller

Go to:

Setup → Apex Classes → New

Paste:

```
public with sharing class StudentController {  
    public Student__c student { get; set; }  
}
```

```

public List<Student__c> studentList { get; set; }

public StudentController() {

    student = new Student__c();

    studentList = [
        SELECT Id, Name, Roll_No__c,
               Class__c, Mobile_No__c
        FROM Student__c
    ];
}

public void saveStudent() {

    insert student;

    student = new Student__c();

    studentList = [
        SELECT Id, Name, Roll_No__c,
               Class__c, Mobile_No__c
        FROM Student__c
    ];
}
}

```

Click:

Save

STEP 5 — Create Visualforce Page

Go to:

Setup → Visualforce Pages → New

Page Name:

StudentPage

Paste:

```

<apex:page controller="StudentController">

    <apex:form>

        <apex:pageBlock title="Student Record Management">

            <apex:pageBlockSection columns="1">

```

```

        <apex:inputField value="{!student.Name}"/>

        <apex:inputField value="{!student.Roll_No__c}"/>

        <apex:inputField value="{!student.Class__c}"/>

        <apex:inputField value="{!student.Mobile_No__c}"/>

    </apex:pageBlockSection>

    <apex:commandButton
        value="Save"
        action="{!saveStudent}"
        rerender="table"/>

</apex:pageBlock>

<apex:pageBlock title="Student List" id="table">

    <apex:pageBlockTable
        value="{!studentList}"
        var="s">

        <apex:column value="{!s.Name}"/>

        <apex:column value="{!s.Roll_No__c}"/>

        <apex:column value="{!s.Class__c}"/>

        <apex:column value="{!s.Mobile_No__c}"/>

    </apex:pageBlockTable>

</apex:pageBlock>

</apex:form>

</apex:page>

```

Click:

Save

STEP 6 — Run Application

Click:

Preview

OR open:

/apex/StudentPage

STEP 7 — Test Application

Enter:

- Name
- Roll No
- Class
- Mobile No

Click:

Save

Verify:

- ✓ record inserted
- ✓ table updated
- ✓ data displayed properly

WHAT TO EXPLAIN TO EXTERNAL

Architecture Flow

```
Visualforce Page
    ↓
Apex Controller
    ↓
Salesforce Database (Custom Object)
```

IMPORTANT VIVA QUESTIONS

What is Apex?

Apex is Salesforce's object-oriented programming language used for backend business logic.

What is Visualforce?

Visualforce is a framework used to design user interfaces in Salesforce.

What is SOQL?

Salesforce Object Query Language used to retrieve records from Salesforce objects.

What is rerender?

rerender refreshes selected components without reloading the complete page.

What is CRUD?

Create, Read, Update, Delete operations.

COMMON MISTAKES

✗ Wrong API field names

Correct:

```
Roll_No__c  
Class__c  
Mobile_No__c
```

✗ Forgetting __c

✗ Saving page before Apex class

✗ Typing object name incorrectly

✗ Not refreshing student list after insert

FINAL RESULT

The application should:

- ✓ accept student details
- ✓ save records in Salesforce database
- ✓ display records dynamically on Visualforce page